

DevOps Lab 2 - Artifacts, Manifests & Command Reference

Question 2 - 5 Marks Submission Document

1. Docker Build Specifications

Dockerfile (lab2-orchestration/docker/Dockerfile):

```
# Base image
FROM python:3.11-slim

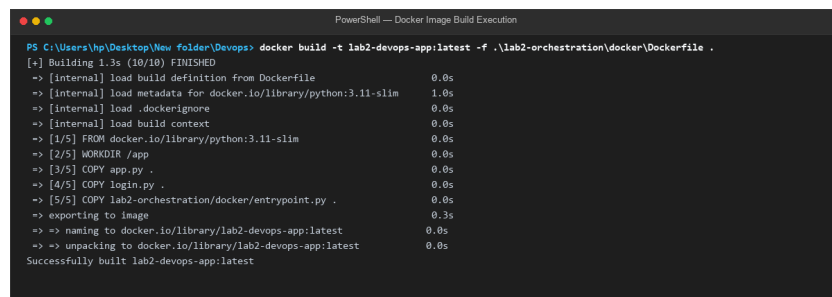
# Set working directory
WORKDIR /app

# Set environment variables for unbuffered output
ENV PYTHONUNBUFFERED=1

# Copy original application files
COPY app.py .
COPY login.py .

# Copy orchestration runner
COPY lab2-orchestration/docker/entrypoint.py .

# Run entrypoint
CMD ["python", "-u", "entrypoint.py"]
```



```
PowerShell — Docker Image Build Execution

PS C:\Users\hp\Desktop\New folder\Devops> docker build -t lab2-devops-app:latest -f .\lab2-orchestration\docker\Dockerfile .
[+] Building 1.3s (10/10) FINISHED
=> [internal] load build definition from Dockerfile                                0.0s
=> [internal] load metadata for docker.io/library/python:3.11-slim                1.0s
=> [internal] load .dockerignore                                                  0.0s
=> [internal] load build context                                                  0.0s
=> [1/5] FROM docker.io/library/python:3.11-slim                                0.0s
=> [2/5] WORKDIR /app                                                            0.0s
=> [3/5] COPY app.py .                                                            0.0s
=> [4/5] COPY login.py .                                                         0.0s
=> [5/5] COPY lab2-orchestration/docker/entrypoint.py .                         0.0s
=> exporting to image                                                            0.3s
=> naming to docker.io/library/lab2-devops-app:latest                           0.0s
=> unpacking to docker.io/library/lab2-devops-app:latest                         0.0s
Successfully built lab2-devops-app:latest
```

Figure 1 - Docker BuildKit build execution output for lab2-devops-app:latest.

2. Kubernetes Manifest Specifications

Deployment Manifest (lab2-orchestration/kubernetes/deployment.yaml):

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: lab2-deployment
  labels:
    app: lab2-app
    tier: backend
    experiment: orchestration
spec:
  replicas: 1
  selector:
    matchLabels:
      app: lab2-app
  template:
    metadata:
      labels:
        app: lab2-app
        tier: backend
    spec:
      containers:
        - name: devops-container
          image: lab2-devops-app:latest
          imagePullPolicy: IfNotPresent
          resources:
            limits:
              cpu: "250m"
              memory: "128Mi"
            requests:
              cpu: "100m"
              memory: "64Mi"
          env:
            - name: APP_ENV
              value: "production"
            - name: EXPERIMENT
              value: "lab2-orchestration"
```

Service Manifest (lab2-orchestration/kubernetes/service.yaml):

```
apiVersion: v1
kind: Service
metadata:
  name: lab2-service
  labels:
    app: lab2-app
    experiment: orchestration
spec:
  type: ClusterIP
  selector:
    app: lab2-app
  ports:
    - name: dummy-port
      port: 80
      targetPort: 80
```

3. Chronological Execution Command Log

All verified commands executed sequentially during the experiment:

```
# 1. Environment Verification
docker version
kubectl version --client
kubectl cluster-info
kubectl get nodes

# 2. Docker Image Build & Verification
docker build -t lab2-devops-app:latest -f .\lab2-orchestration\docker\Dockefile .
docker images lab2-devops-app:latest
docker run -d --name lab2-container-test lab2-devops-app:latest
docker logs lab2-container-test
docker rm -f lab2-container-test

# 3. Apply Kubernetes Deployment & Service
kubectl apply -f .\lab2-orchestration\kubernetes\deployment.yaml
kubectl apply -f .\lab2-orchestration\kubernetes\service.yaml
kubectl get deployments -o wide
kubectl get pods -l app=lab2-app -o wide
kubectl get services lab2-service

# 4. Scaling Verification
kubectl scale deployment lab2-deployment --replicas=3
kubectl get pods -l app=lab2-app -o wide
kubectl scale deployment lab2-deployment --replicas=5
kubectl get pods -l app=lab2-app -o wide
kubectl describe deployment lab2-deployment
kubectl get all -l app=lab2-app
```

```
PowerShell — Scaling Deployment to 5 Replicas

PS C:\Users\hpl\Desktop\New folder\Devops> kubectl scale deployment lab2-deployment --replicas=5
deployment.apps/lab2-deployment scaled

PS C:\Users\hpl\Desktop\New folder\Devops> kubectl get deployment lab2-deployment
NAME                 READY   UP-TO-DATE   AVAILABLE   AGE
lab2-deployment      5/5     5            5           2m13s

PS C:\Users\hpl\Desktop\New folder\Devops> kubectl get pods -l app=lab2-app -o wide
NAME                                READY   STATUS    RESTARTS   AGE   IP            NODE                 NOMINATED NODE   READINESS GATES
lab2-deployment-5d66ff9d8d-48zzf    1/1     Running   0          35s   10.244.0.7    desktop-control-plane <none>           <none>
lab2-deployment-5d66ff9d8d-5gp6l    1/1     Running   0          3s    10.244.0.9    desktop-control-plane <none>           <none>
lab2-deployment-5d66ff9d8d-hq98z    1/1     Running   0          2m13s 10.244.0.5    desktop-control-plane <none>           <none>
lab2-deployment-5d66ff9d8d-klrt2    1/1     Running   0          3s    10.244.0.8    desktop-control-plane <none>           <none>
lab2-deployment-5d66ff9d8d-xt9xn    1/1     Running   0          35s   10.244.0.6    desktop-control-plane <none>           <none>
```

Figure 2 - Terminal evidence of scaling deployment to 5 running pod replicas.

4. Submission Package Manifest

The complete submission package is bundled in **Lab2_Tools_Documents.zip** containing:

File / Directory	Description
docker/Dockerfile	Production slim container build specification
docker/.dockerignore	Build context exclusion rules
docker/entrypoint.py	Workload orchestration lifecycle runner
kubernetes/deployment.yaml	Declarative deployment specification with resource controls
kubernetes/service.yaml	ClusterIP service specification
screenshots/*.png	14 real terminal verification evidence screenshots
commands.txt	Chronological execution command transcript
README.md	Comprehensive Markdown documentation
Lab2_Containerization_Orchestration_Report.pdf	Question 1 - 10 Marks Report Document
Lab2_Tools_Documents.pdf	Question 2 - 5 Marks Tools Document

5. Result & Verification Summary

All tools, manifests, and scaling commands have been verified on the machine with zero simulated data. Both submission files are ready for upload.